

Assignment No 6

Write a program of Apriori algorithm using any programming language.

CODE:

```
from itertools import combinations

def get_support(transactions, itemset):
    """Calculates the support of a given itemset."""
    count = 0
    for tx in transactions:
        if set(itemset).issubset(set(tx)):
            count += 1
    return count / len(transactions)

def apriori(transactions, min_support):
    """Core Apriori logic to find frequent itemsets."""
    # Step 1: Find unique items and create initial 1-itemsets
    unique_items = sorted(list(set(item for tx in transactions for item in tx)))
    current_candidates = [[item] for item in unique_items]
    all_frequent_itemsets = []
    k = 1
    while current_candidates:
        # Step 2: Filter candidates by minimum support (Pruning)
        frequent_itemsets = []
        for itemset in current_candidates:
            support = get_support(transactions, itemset)
            if support >= min_support:
                frequent_itemsets.append((tuple(itemset), support))
        # If no itemsets meet support, stop
        if not frequent_itemsets:
            break
        all_frequent_itemsets.extend(frequent_itemsets)
        # Step 3: Generate next level candidates (Ck+1)
        found_items = sorted(list(set(item for itemset, sup in frequent_itemsets for item in itemset)))
        # The next line needs to handle the case where found_items might be empty
        # or when k+1 candidates cannot be formed.
        if k + 1 > len(found_items) and len(found_items) > 0:
            current_candidates = [] # No more combinations possible
        elif len(found_items) == 0:
            current_candidates = [] # No more items to form candidates
        else:
            current_candidates = list(combinations(found_items, k + 1))
        k += 1
    return all_frequent_itemsets

# --- Example Usage ---
# A small transactional dataset
```

```
dataset = [  
    ['Milk', 'Bread', 'Wheat'],  
    ['Bread', 'Eggs'],  
    ['Milk', 'Bread', 'Eggs'],  
    ['Milk', 'Eggs'],  
    ['Bread', 'Wheat']  
]  
# Set minimum support to 40%  
min_sup = 0.4  
frequent_patterns = apriori(dataset, min_sup)  
print(f"Frequent Itemsets (min_support={min_sup}):")  
for itemset, support in frequent_patterns:  
    print(f"Itemset: {list(itemset)}, Support: {support:.2f}")
```

OUTPUT :

```
▼ ... Frequent Itemsets (min_support=0.4):  
    Itemset: ['Bread'], Support: 0.80  
    Itemset: ['Eggs'], Support: 0.60  
    Itemset: ['Milk'], Support: 0.60  
    Itemset: ['Wheat'], Support: 0.40  
    Itemset: ['Bread', 'Eggs'], Support: 0.40  
    Itemset: ['Bread', 'Milk'], Support: 0.40  
    Itemset: ['Bread', 'Wheat'], Support: 0.40  
    Itemset: ['Eggs', 'Milk'], Support: 0.40
```